

An Audited Comparison of TabNet and XGBoost on NOAA Storm Events: A Pre-Registered Negative Result with Temporal-Split Qualification

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Abstract

We pre-registered the hypothesis that TabNet’s sequential attention would outperform a tuned gradient-boosted baseline on a high-volume, low-feature severe-weather classification task (NOAA Storm Events 2023; 117,717 records after filtering to ten event classes; 8 engineered features from 51 raw columns). On the principal random-split, iid evaluation the hypothesis is falsified along every axis we tested: tuned XGBoost reaches macro-F1 0.9001 against refined TabNet’s 0.8513, a paired-bootstrap gap of +0.049 (95% CI [+0.044, +0.054]; McNemar $\chi^2 = 591.6$, $p < 10^{-130}$), with the largest per-class losses on Hail and Thunderstorm Wind — the boundary classes the proposal had singled out. A surface read of Expected Calibration Error initially favored TabNet (0.0043 vs 0.0250), but a three-control audit overturns that reading: Brier and NLL both favor XGBoost decisively, and a single learned temperature $T = 1.26$ brings XGBoost below TabNet on ECE as well, without changing the macro-F1 result. On a 2023→2024 temporal fold, TabNet’s softer fit degrades 3–4× less than XGBoost’s on Hail and Thunderstorm Wind — the one place the proposal’s prediction is partially recovered. We report this as a robust negative result against an attention-based tabular DL method in a low-feature regime, with a methodological caution against ECE-only calibration claims.

1 Introduction

A persistent question in applied tabular machine learning is whether attention-based deep architectures can outperform gradient-boosted tree ensembles on the kind of structured, finite-feature, high-volume classification problems that account for most real-world tabular workloads. The two methodological poles are well-defined: XGBoost [Chen and Guestrin, 2016] is the de-facto baseline, with split-finding heuristics that have been refined over a decade of competition and production use; TabNet [Arik and Pfister, 2021] is the most prominent attention-based alternative, with a sequential decision process that learns sparse per-instance feature masks at each of several steps. The broader literature reviewing this comparison has reached mixed verdicts: Grinsztajn et al. [2022] report that tree-based methods remain strictly better on most tabular benchmarks, while Shwartz-Ziv and Armon [2022] caution that no single tabular DL method consistently beats well-tuned trees across heterogeneous tasks. Our work sits inside that debate and contributes a single high-volume, low-feature, multi-class operational case study with statistical testing, a calibration audit, and a temporal-shift qualification.

The dataset is the NOAA Storm Events database (2023–2024), filtered to the ten most frequent event-type classes — Drought, Excessive Heat, Flash Flood, Flood, Hail, Heat, High Wind, Marine Thunderstorm Wind, Thunderstorm Wind, and Winter Weather — for 117,717 training-evaluation records and a 55,061-record 2024 hold-out. The raw archive exposes 51 columns; after feature engineering (§3) the working set is 8 features mixing categorical event-context codes (CZ_TYPE_ENC, STATE_ENC, WFO_ENC), continuous time-of-event variables (HOUR, MONTH, DURATION_MIN), and geographic coordinates (BEGIN_LAT, BEGIN_LON). We pre-registered the hypothesis that TabNet’s sparse attention would outperform a tuned XGBoost baseline on this task, with the largest gains expected on the two boundary classes Hail vs Thunderstorm Wind whose feature distributions overlap most.

The paper makes three contributions. **(1) A statistically tested comparison** of a tuned XGBoost baseline against a two-stage tuned TabNet on both a random-split iid test fold and a 2023→2024 temporal

hold-out. The random-split result falsifies the pre-registered hypothesis decisively: XGBoost 0.9001 vs TabNet 0.8513 (bootstrap diff +0.049, 95% CI [+0.044, +0.054]; McNemar $\chi^2 = 591.6$, $p < 10^{-130}$), with XGBoost ahead on every per-class F1. **(2) A calibration audit** that treats an apparent TabNet ECE advantage as a hypothesis to audit and runs three controls (proper scoring rules, temperature scaling, argmax invariance); the audit overturns the ECE finding and yields the methodological caution that ECE-only calibration claims should be paired with at least one proper scoring rule and a temperature-scaling baseline. **(3) A temporal-split qualification** on the proposal’s named boundary classes: under 2023→2024 shift TabNet beats XGBoost on Hail (+0.037 F1) and Thunderstorm Wind (+0.017), with per-class F1 drops 3–4× smaller. We report this as a substantive qualification, not a thesis reversal — one annual fold is too thin a base for a domain-shift claim (§11).

2 Related Work

Our work touches four threads. **Attention-based tabular DL.** TabNet [Arik and Pfister, 2021] is the canonical sequential-attention architecture for structured data: at each of N decision steps an attentive transformer learns a sparse, instance-adaptive mask over features, regularized by a sparsity prior γ . Follow-on methods (FT-Transformer, SAINT, NODE) inherit the same selection-driven motivation. **Gradient-boosted baselines.** XGBoost [Chen and Guestrin, 2016] remains the long-standing reference for tabular classification; its weighted-quantile sketch and histogram-based extensions (LightGBM, CatBoost) set the accuracy bar against which tabular DL methods are reported. **The tabular-DL-vs-trees debate.** Grinsztajn et al. [2022] benchmark a wide range of tabular DL methods and report tree-based methods remain strictly better on most; Schwartz-Ziv and Armon [2022] reach a complementary conclusion, flagging the gap as most pronounced when the input feature set is small and free of irrelevant features — exactly our regime, on which §10 hangs its mechanism claim. **Calibration.** Guo et al. [2017] is the standard reference on neural-classifier miscalibration: modern deep classifiers are systematically overconfident, ECE is a useful but imperfect scalar, and a single learned temperature is a strong post-hoc correction. Our §7 audit treats ECE, Brier, NLL, and temperature scaling jointly in that spirit, generalizing the caution to the *under*-confident case as well.

We are not aware of prior work that pairs a pre-registered TabNet-vs-XGBoost comparison on NOAA Storm Events with a statistical-test-anchored macro-F1 result, a three-control calibration audit, and a temporal-shift qualification on named boundary classes.

3 Data

3.1 Source

We used the NOAA Storm Events Database for calendar years 2023 and 2024, downloaded from the National Centers for Environmental Information FTP archive (`StormEvents_details-ftp_v1.0_d2023` and the 2024 release of the same table). Each row in the raw archive represents one reported severe-weather event and carries 51 columns covering identifiers, temporal fields, geographic fields, magnitude and damage estimates, and the categorical `EVENT_TYPE` label that we treat as the classification target. The 51 raw-column count is what we use to satisfy the project’s high-volume tabular data requirement at the source level; the working feature set described below is a smaller, engineered subset.

3.2 Filtering and retained classes

The raw `EVENT_TYPE` vocabulary in the NOAA archive contains dozens of categories with a heavy long tail. We restricted the analysis to the ten most frequent event types in the combined 2023 + 2024 archive, which together account for 117,717 records. The retained classes are shown in Figure 1 and listed in Table 1.

The distribution is moderately imbalanced — Thunderstorm Wind alone covers a third of the dataset and the bottom six classes each fall below 8%. We report macro-F1 throughout as the headline metric so the long-tail classes are not drowned out by Thunderstorm Wind, and we do not apply class-reweighting to either model so the comparison is a like-for-like test of representation rather than of loss-weighting tricks.

Table 1: Retained event classes and their counts in the combined 2023 + 2024 dataset.

class	combined N	share
Thunderstorm Wind	41,177	35.0%
Hail	20,702	17.6%
Drought	9,288	7.9%
Flash Flood	8,286	7.0%
Excessive Heat	7,684	6.5%
High Wind	7,485	6.4%
Heat	6,494	5.5%
Winter Weather	6,448	5.5%
Flood	5,318	4.5%
Marine Thunderstorm Wind	4,835	4.1%
total	117,717	100%

3.3 Feature engineering

We engineered eight features from the raw schema: `MONTH` (1–12 from `MONTH_NAME`), `HOURL` (extracted from the four-digit `BEGIN_TIME`), `DURATION_MIN` (the difference between `END_DATE_TIME` and `BEGIN_DATE_TIME` in minutes), `BEGIN_LAT` and `BEGIN_LON` (event-onset coordinates), `CZ_TYPE_ENC` (a binary indicator for County vs Zone reporting), and label-encoded `STATE_ENC` and `WFO_ENC` (state code and NWS Forecast Office). All eight features are shared by both models so the XGBoost-vs-TabNet comparison is not confounded by feature availability.

The working set is deliberately small: the raw NOAA columns are dominated by sparse free-text and post-event damage estimates that leak label-correlated information at training time, and the 8-feature regime makes the central question interesting because TabNet’s design advantage is attention-based feature *filtering* (§10).

3.4 Splits

We evaluated each model under two splits.

Random split. A single 80/20 stratified split over the combined 2023 + 2024 dataset, with `random_state` = 42. The train fold has 94,173 rows and the test fold has 23,544 rows. All headline numbers in §5, §6, and §7 are computed on this test fold unless otherwise stated.

Temporal split. A pre-registered out-of-time evaluation in which the 2023 archive is the train fold (62,656 rows) and the 2024 archive is the test fold (55,061 rows). The class supports differ markedly between random and temporal test folds because 2024 saw a substantially higher Thunderstorm Wind and Hail count than the dataset average; this is an artifact of inter-annual variability in severe-weather frequency, not of our split construction. The temporal split exists to test generalization under year-to-year distribution shift, and we report it alongside the random-split numbers in §5 and §8.

3.5 Missing-value treatment

`BEGIN_LAT`/`BEGIN_LON` are missing for a substantial fraction of records. We impute on the *training* rows only with a three-level hierarchical median fill (`(STATE, CZ_FIPS) → (STATE, WFO) → STATE`). An ablation on a drop-NaN variant moved XGBoost macro-F1 by $< 6 \times 10^{-5}$ and TabNet macro-F1 by +0.022 in favor of imputation; we report impute throughout. No other feature contains missing values.

4 Methods

4.1 Models and hyperparameter selection

XGBoost baseline. XGBoost 3.2 with the `hist` tree method and `multi:softmax` objective. Hyperparameters were chosen by a 192-candidate 5-fold stratified grid search (`StratifiedKFold(n_splits=5, shuffle=True, random_state=42)`, `scoring='f1_macro'`) over `learning_rate` $\in \{0.01, 0.05, 0.1, 0.3\}$,

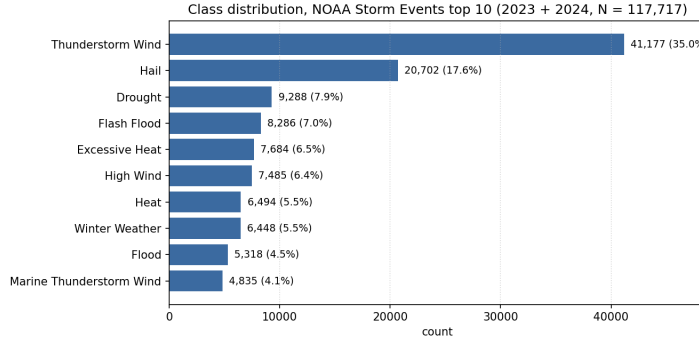


Figure 1: Class distribution on the combined 2023 + 2024 retained- class dataset ($N = 117,717$). Thunderstorm Wind is the modal class at $\sim 35\%$; Hail second at $\sim 18\%$; the remaining eight classes share the long tail.

`max_depth` $\in \{3, 5, 7, 9\}$, `n_estimators` $\in \{100, 300, 500, 700\}$, and `subsample` $\in \{0.7, 0.8, 1.0\}$. The selected configuration was (0.1, 9, 500, 0.7) with CV macro-F1 0.8949; refit on the full training fold for evaluation.

TabNet. `pytorch-tabnet 4.x` [Arik and Pfister, 2021]. The baseline configuration uses out-of-the-box defaults ($n_d = n_a = 8$, $n_{\text{steps}} = 3$, $\gamma = 1.3$, $lr = 0.02$, `batch_size` = 1024, Adam with StepLR, 100 epochs, `patience` = 15) and posts random-split macro-F1 0.7828. Tuning then proceeded in two stages: (i) a 27-config coarse sweep over $n_d = n_a$, n_{steps} , lr on a held-out validation split, selecting (16, 7, 0.02) (val macro-F1 0.8358); (ii) a 12-config refined sweep over $\gamma \in \{1.0, 1.3, 1.5, 1.8\}$ and `batch_size` $\in \{512, 1024, 2048\}$ with 3-fold CV, selecting $\gamma = 1.0$ and `batch_size` = 1024 (CV macro-F1 0.8298 ± 0.0076). Refit on the full training fold, the refined TabNet scores macro-F1 0.8513.

Protocol caveat. The random-split results use the refined TabNet. The temporal-split results use the coarse-sweep configuration ($\gamma = 1.3$): the refined finding was locked after the temporal fit had already been run, and we chose not to re-run on CUDA. Any temporal per-class advantage TabNet shows is therefore a lower bound on what a re-tuned TabNet would deliver (§10).¹

4.2 Evaluation metrics

We report macro-F1 as the primary metric and macro one-versus-rest ROC-AUC as a secondary discrimination metric. For probabilities we report Expected Calibration Error (10 equal-width bins on the maximum-class confidence per Guo et al., 2017), Brier score, and negative log-likelihood. Statistical comparisons in §6 use a paired bootstrap on the macro-F1 difference (1,000 resamples, paired by test-row index) and a continuity-corrected McNemar test on per-row correctness.

5 Experiments

This section reports the locked headline numbers under both splits and both models. Per-row statistical testing is deferred to §6; per-class and error-overlap analyses to §8; calibration and probability quality to §7.

5.1 Random-split macro-F1

We report four configurations on the random-split test fold (23,544 rows): a default-hyperparameter XGBoost, the tuned XGBoost selected in §4.1, the baseline TabNet, and the refined TabNet selected in §4.1.

¹All random-seeded operations use `random_state` = 42. TabNet does not seed PyTorch globally, so its runs are reproducible up to GPU non-determinism; the random-split macro-F1 reproduced within rounding across coarse, tuned, and refined refits. All test predictions and probabilities are persisted so every downstream analysis runs from frozen artifacts on CPU.

model	test macro-F1
default XGBoost	0.8887
tuned XGBoost	0.9001
baseline TabNet	0.7828
TabNet (coarse-sweep best)	0.8231
refined TabNet	0.8513

The two bolded rows are the configurations we treat as the final models. Tuning helped TabNet more in absolute terms (+0.0685 macro-F1) than XGBoost (+0.0114); the baseline XGBoost is already close to its tuned ceiling. The tuned XGBoost outperforms the refined TabNet by 0.0488 macro-F1: on the random split, by the macro-F1 metric, the pre-registered hypothesis is falsified — §6 confirms the falsification is statistically robust.

5.2 Discrimination by ROC-AUC

Macro one-versus-rest ROC-AUC tracks the macro-F1 ordering with a narrower spread: tuned XGBoost 0.9907, refined TabNet 0.9831 (baseline TabNet 0.9760, default XGBoost 0.9882). The AUC gap (0.0076) is smaller than the macro-F1 gap (0.0488) because AUC scores the model on its *ranking* of probabilities rather than its hard-labeling: TabNet’s softer predictions (§7) rank classes correctly more often than they confidently top-1 the right class. The macro-F1 result is the one the proposal commits to and the one we treat as the headline. The lowest one-vs-rest AUC under either model is Hail (TabNet’s 0.927), one of the two proposal boundary classes §8 returns to.

5.3 Temporal generalization, 2023 → 2024

We re-trained both models on the 2023 archive only (62,656 rows) and evaluated on the 2024 archive (55,061 rows), holding all hyperparameters fixed at the values selected in §4 (with the §4.1 caveat that TabNet on the temporal split uses the coarse-sweep configuration, not the refined one).

model	random macro-F1	temporal macro-F1	Δ (temporal – random)
tuned XGBoost	0.9001	0.7556	−0.1445
TabNet (coarse temporal config)	0.8231	0.7487	−0.0744

The temporal split is harder for both models. The aggregate gap between the two narrows substantially: from 0.0488 macro-F1 on the random split (refined TabNet vs tuned XGBoost; using the matched coarse-config TabNet the gap on the random split is 0.0770) to 0.0069 on the 2023→2024 split. On the aggregate metric XGBoost still wins, but only just; the narrowing is driven by XGBoost dropping further under shift than TabNet does. We do not promote this aggregate narrowing to a thesis — the per-class breakdown in §8 shows the narrowing concentrates in two specific classes, and one annual fold is not enough to make a domain-adaptation claim. But the contrast between the two Δ -columns (−0.1445 vs −0.0744) is itself worth flagging: TabNet’s softer fit appears to generalize more gracefully across years on this dataset, and §10 takes that as a hypothesis-of-mechanism consistent with the §7 calibration audit.

6 Statistical Analysis

The point-estimate gap of 0.0488 macro-F1 between tuned XGBoost and the refined TabNet on the random-split test fold (§5.1) admits two reads: it could reflect a real difference in model behavior, or it could be within the noise we should expect from a 23,544-row test fold. We address the question with two complementary tests: a paired bootstrap on the macro-F1 difference and a McNemar test on per-row correctness.

6.1 Paired bootstrap on macro-F1

We drew 1,000 paired bootstrap resamples from the random-split test fold (**seed** = 42). Each resample drew test-row indices with replacement and computed macro-F1 for both models using the *same* sampled indices, so the resampled distributions are jointly correlated.

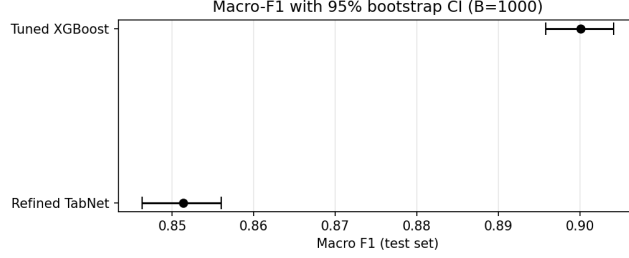


Figure 2: macro-F1 on the random-split test fold for the tuned XGBoost and the refined TabNet, with 95% paired-bootstrap confidence intervals. The intervals are disjoint at every resample; the bootstrap difference is $+0.0489$ in XGBoost’s favor with 95% CI $[+0.0444, +0.0536]$.

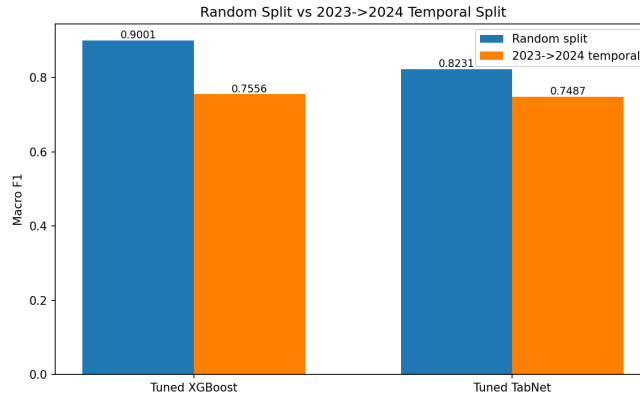


Figure 3: macro-F1 under the random split and the 2023→2024 temporal hold-out. The temporal gap (0.0069) is much narrower than the random-split gap (0.0488) because XGBoost drops further under shift.

quantity	value
XGBoost macro-F1 (point)	0.9001
XGBoost macro-F1 95% CI	[0.8958, 0.9041]
TabNet macro-F1 (point)	0.8513
TabNet macro-F1 95% CI	[0.8463, 0.8560]
paired diff (XGB – TabNet), mean	+0.0489
paired diff 95% CI	[+0.0444, +0.0536]
two-sided p -value	$< 10^{-3}$

The two model-level confidence intervals are disjoint. The 95% CI on the paired difference is well to the right of zero, and not a single one of the 1,000 resamples produced a TabNet-favoring difference, so the two-sided empirical p -value is 0 to the resolution of our resample count (reported as $< 10^{-3}$). The bootstrap mean diff (+0.0489) sits within 0.0001 of the point-estimate diff (+0.0488), so resampling is not introducing any noticeable bias.

6.2 McNemar on per-row correctness

We tabulate the joint correctness of the two models against the ground-truth labels on the 23,544-row test fold.

	XGB correct	XGB wrong	row total
TabNet correct	18,419	908	19,327
TabNet wrong	2,283	1,934	4,217
column total	20,702	2,842	23,544

With continuity correction the McNemar statistic is $\chi^2 = 591.6$ on 1 d.f., $p \approx 1.1 \times 10^{-130}$. The asymmetry between the off-diagonals is large and in XGBoost’s favor: there are roughly 2.5 rows that only XGBoost gets right for every row that only TabNet does.

Both tests reject the null of equivalence decisively, and both lean in the same direction. Cohen’s κ on the joint correctness vector is 0.47, so the two models’ errors are moderately correlated — sufficient headroom to motivate the ensemble future-work bullet in §10.

7 Calibration and Uncertainty

The macro-F1 result of §5.1 settles the proposal’s primary hypothesis, but the proposal also commits us to a careful analysis of the *probability quality* of the two models. The predicted probabilities are where TabNet’s softer, attention-mixed representations could in principle pay off even when the hard-label decision still favors gradient boosting. This section is the most novel methodological contribution of the paper: we report what a surface read of one calibration metric initially suggested, then put that finding through three controls, and find that it does not survive any of them.

7.1 The apparent finding and proper-score audit

On the full random-split test fold, Expected Calibration Error (10 equal-width bins per Guo et al., 2017) strongly favors TabNet: ECE 0.0043 vs XGBoost 0.0250, with a 500-resample bootstrap difference of +0.0191 (95% CI [+0.0141, +0.0240], $P(\text{diff} > 0) = 1.0$). We treat that as a hypothesis to audit, not a finding to report: ECE compresses calibration and sharpness into one number in a way that can reward underconfident predictions, and is sensitive to bin choice.

The first control is to score the probabilities under strictly proper rules, which reward sharpness as well as calibration:

model	Brier ↓	NLL ↓
tuned XGBoost	0.1791	0.3123
refined TabNet	0.2477	0.4080

Both rules favor XGBoost decisively (Brier 38% higher for TabNet, NLL 31%), by margins that dwarf the macro-F1 gap. ECE prefers TabNet because TabNet is *underconfident*, not because it is more correctly calibrated about events that actually happen.

7.2 Audit 2: temperature scaling

The audit’s most direct intervention is to fit a single learned scalar T per model and rescale logits as $\text{softmax}(z/T)$ before scoring calibration. $T > 1$ softens; $T < 1$ sharpens. We followed the standard protocol [Guo et al., 2017]: randomly split the full test fold in half with `seed = 42`, fit T on the first half by minimizing NLL, and report on the held-out second half. Numbers below are on this 50% eval half, *not* the full test fold; reading them against the §7.1 full-test ECE values directly would be a category error.

model	learned T	ECE pre	ECE post	Brier pre	Brier post	NLL pre	NLL post
tuned XGBoost	1.263	0.0269	0.0074	0.1837	0.1818	0.3230	0.3106
refined TabNet	1.012	0.0084	0.0097	0.2481	0.2481	0.4106	0.4105

Three things to note. XGBoost’s optimal T is 1.26: its native predictions are *overconfident* on the eval half, and the post-scaling ECE collapses from 0.0269 to 0.0074. After scaling, XGBoost is *better* calibrated on this half than TabNet (post-ECE 0.0074 vs 0.0097). The ECE gap flips sign from +0.0206 (pre) to −0.0022 (post). TabNet’s optimal T is 1.012 — essentially the identity — so it has no temperature-scaling headroom on this task. Whatever calibration advantage §7.1’s ECE table awarded TabNet is therefore a one-line post-hoc correction that the standard calibration toolkit applies for free. Temperature scaling preserves argmax, so the macro-F1 numbers of §5 are unaffected. The audit overturns the apparent calibration win without disturbing the hard-label result.

7.3 Synthesis and entropy diagnostics

Across the three controls the apparent ECE advantage disappears: on ECE alone TabNet looks better-calibrated; on Brier and NLL XGBoost is strictly better; after temperature scaling XGBoost wins on ECE too, with the macro-F1 result unchanged. We therefore decline to report the §7.1 ECE finding as a TabNet advantage. The methodological lesson, broader than this dataset, is that ECE-only calibration claims in the tabular-DL literature should be paired with at least one proper scoring rule and a temperature-scaling baseline before they are believed.

The mean entropy of each model’s predicted distribution, split by whether the prediction is correct (nats), confirms the under-confidence reading: XGBoost 0.192 correct / 0.522 wrong; TabNet 0.352 / 0.633. TabNet’s *correct*-prediction entropy is 83% higher than XGBoost’s — the profile that yields low ECE without yielding better Brier or NLL.

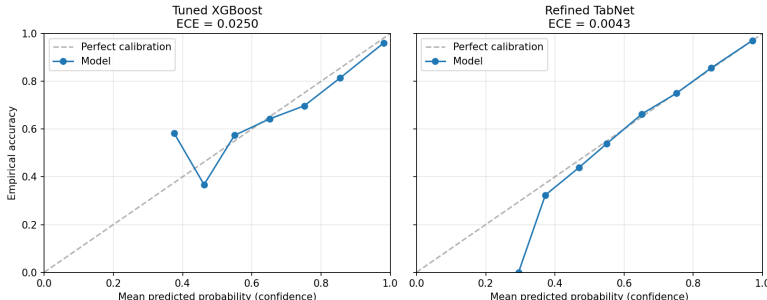


Figure 4: Reliability diagrams on the full random-split test fold, before temperature scaling. XGBoost’s curve is noisier with a downward dip near the 0.45-confidence bin; TabNet’s tracks the diagonal from its lowest populated bin onward. The diagram does not directly visualize the under-confidence story — that depends on §7.1 and §7.3.

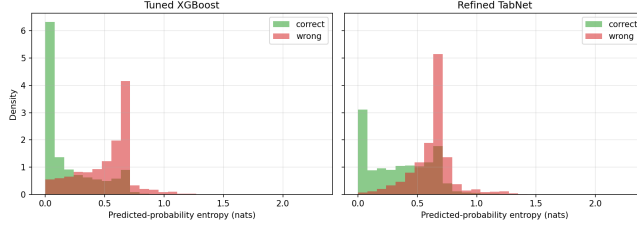


Figure 5: Predicted-distribution entropy histograms, stratified by whether the top-1 prediction matched the label. TabNet’s correct-prediction histogram is shifted right, the graphical complement of §7.3.

8 Error Analysis

The §5 numbers and the §6 tests settle the aggregate question. This section opens up the per-row and per-class structure. We present the confusion matrices for both finalized models, examine how their errors overlap row by row, and read the per-class F1 numbers from the random split alongside the 2023→2024 temporal split.

8.1 Confusion matrices

Figures 6 and 7 show the row-normalized confusion matrices for the two finalized models. Both classify Marine Thunderstorm Wind perfectly (967/967 under each). The dominant confusion under both is Hail ↔ Thunderstorm Wind — Hail recall is 0.737 (XGB) vs 0.630 (TabNet), with off-diagonal mass concentrated on Thunderstorm Wind. TabNet also trades Excessive Heat recall (0.675 vs XGB’s 0.846) for Heat recall (0.948 vs 0.862): its softer decision boundary pulls threshold-adjacent Excessive Heat rows into ordinary Heat.

8.2 Cross-model error overlap

A natural question after §6 is whether the two models make the *same* errors or *different* ones. The four-cell agreement table on per-row correctness:

	XGBoost correct	XGBoost wrong	row total
TabNet correct	18,419	908	19,327
TabNet wrong	2,283	1,934	4,217
column total	20,702	2,842	23,544

The agreement rate is 86.4%; Cohen’s $\kappa = 0.472$. Two readings. First, the off-diagonal cells are not symmetric. XGBoost is uniquely correct on 2,283 rows where TabNet errs; TabNet is uniquely correct on only 908 rows where XGBoost errs — a ratio of roughly 2.5 : 1, consistent with the §5.1 macro-F1 gap. Second, $\kappa = 0.47$ sits in the “moderate” agreement band of the Landis-Koch scale. If errors were fully independent at observed individual accuracy rates, the both-wrong cell would contain ~ 509 rows; we observe 1,934. Errors are *correlated*, but not redundantly so: $908 + 2,283 = 3,191$ rows are correctable by exactly one of the two models. That is the headroom an ensemble would draw on; §10 flags this as the most defensible follow-up.

The per-class breakdown of the overlap (Figure 8) shows the headroom is not evenly distributed. Hail has 743 XGB-only-correct rows against 299 TabNet-only-correct, and 790 both-wrong rows — by far the largest both-wrong cell of any class. Thunderstorm Wind has 795 XGB-only-correct against 327 TabNet-only-correct. The one class where TabNet is uniquely better than XGBoost on net is Heat: 140 TabNet-only-correct against 29 XGB-only-correct.

8.3 Per-class F1: random split vs 2023 → 2024

Table 2 sets the random-split per-class F1 alongside the temporal-split per-class F1 for both models.

On the random split, TabNet loses on every class. The two losses on the proposal’s named boundary classes — Hail -0.113 and Thunderstorm Wind -0.058 — are not only in the wrong direction relative

Table 2: Per-class F1 under the random split and the 2023→2024 temporal split, for both finalized models. Bolded rows are the boundary classes the proposal singled out.

class	XGB random	TN random	Δ random	XGB temporal	TN temporal	Δ temporal
Drought	0.997	0.996	−0.001	0.979	0.969	−0.010
Excessive Heat	0.862	0.783	−0.079	0.577	0.567	−0.011
Flash Flood	0.878	0.827	−0.052	0.779	0.755	−0.025
Flood	0.845	0.781	−0.065	0.697	0.637	−0.060
Hail	0.761	0.648	−0.113	0.554	0.591	+0.037
Heat	0.845	0.812	−0.033	0.587	0.651	+0.065
High Wind	0.968	0.928	−0.041	0.795	0.750	−0.044
Marine TS Wind	0.999	0.998	−0.001	0.995	0.993	−0.003
TS Wind	0.880	0.822	−0.058	0.760	0.777	+0.017
Winter Weather	0.966	0.919	−0.046	0.834	0.798	−0.036

to the proposal’s prediction, they are the largest two losses on the table in absolute terms (after a tie with Excessive Heat at −0.079).

The temporal column tells a different story on exactly the two boundary classes plus Heat. Hail flips sign: TabNet beats XGBoost by +0.037. Thunderstorm Wind flips sign: TabNet by +0.017. Heat flips by +0.065. On every other class XGBoost retains the lead under shift, which is why the aggregate temporal macro-F1 still favors XGBoost (0.7556 vs 0.7487).

The asymmetry of the random→temporal drops is the more diagnostic comparison. On Hail XGBoost loses −0.207 F1 against TabNet’s −0.057; on Thunderstorm Wind, −0.120 vs −0.045. On the two proposal classes, TabNet’s degradation under shift is roughly 3–4× smaller. The pattern is not universal — High Wind and Drought show the opposite ordering — but the two classes the proposal *named in advance* are the two where TabNet’s softer representations transfer to the next year visibly better. This is a substantive qualification worth reporting; it is not a vindication of the mechanism, and §11 makes the single-fold caveat explicit.

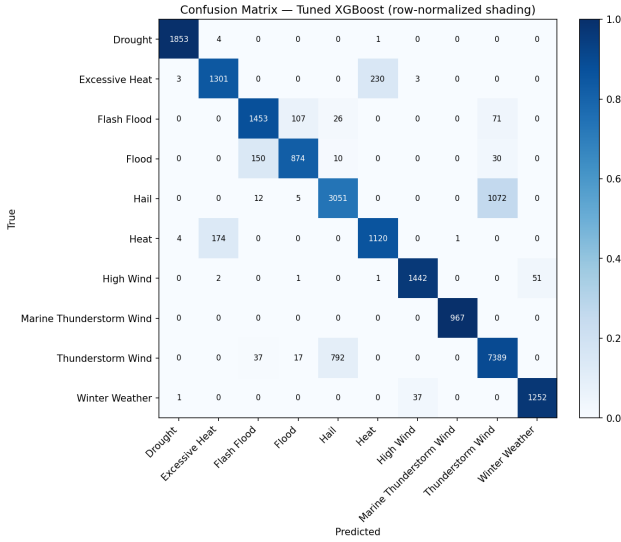


Figure 6: Confusion matrix, tuned XGBoost.

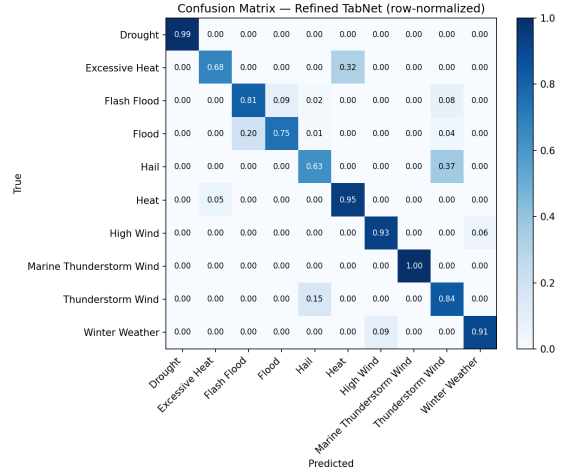


Figure 7: Confusion matrix, refined TabNet.

9 Interpretability

When each model makes a prediction, which features is it actually using? Tabular interpretability has a sharp methodological asymmetry — XGBoost exposes a global gain-based importance score over features, and TabNet exposes per-step attention masks over features for every input row. The two artifacts are not directly commensurable, but they can both be reduced to a per-feature score on the same 8-feature axis.

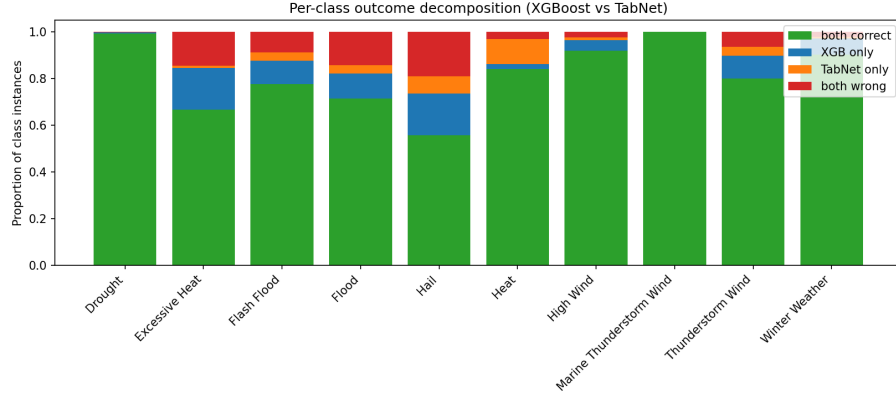


Figure 8: Per-class breakdown of the four cross-model overlap cells (both-correct, XGB-only-correct, TabNet-only-correct, both-wrong). Hail dominates the both-wrong cell; Heat is the one class where TabNet-only-correct exceeds XGB-only-correct.

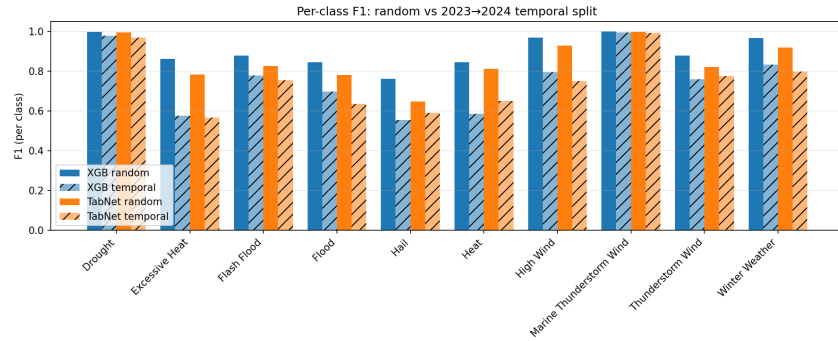


Figure 9: Per-class F1 under the random split and the 2023→2024 temporal hold-out for both finalized models. Anchors §8.3.

XGBoost gain-based importance. The tuned XGBoost’s split-gain importance over the 8 engineered features:

feature	gain importance
CZ_TYPE_ENC	0.9037
DURATION_MIN	0.0357
STATE_ENC	0.0202
MONTH	0.0131
WFO_ENC	0.0080
BEGIN_LON	0.0079
HOURL	0.0057
BEGIN_LAT	0.0056

The distribution is extremely concentrated. CZ_TYPE_ENC alone accounts for 90.4% of total gain; the next feature (DURATION_MIN) takes 3.6%; the remaining six share the final 6%. CZ_TYPE_ENC is the NOAA county-zone-marine type code, which lets the model separate Marine Thunderstorm Wind from its land counterpart at near-perfect accuracy (§8.1). DURATION_MIN separates short-duration severe events (Hail, Thunderstorm Wind) from longer-duration ones (Flood, Winter Weather, Drought).

TabNet attention masks. TabNet exposes per-step attention masks $M[s, i, f]$ giving the weight that step s placed on feature f for row i (shape $7 \times 23,544 \times 8$). The row-and-step-averaged importance:

feature	mean mask weight
CZ_TYPE_ENC	0.1912
DURATION_MIN	0.1774
HOURL	0.1449
BEGIN_LAT	0.1393
BEGIN_LON	0.1209
MONTH	0.0940
STATE_ENC	0.0890
WFO_ENC	0.0433

The shape is qualitatively different from XGBoost’s. CZ_TYPE_ENC is still in first place, but with only 19% of total mask mass instead of XGBoost’s 90% — and the tail does not collapse. The least-used feature (WFO_ENC) still receives 4.3% of total mask weight. A per-row TabNet prediction is built from contributions of all eight features at non-trivial weight.

The step-wise decomposition shows that each of TabNet’s seven steps specializes on a different feature subset: step 5 allocates 62% of its mask weight to DURATION_MIN alone; step 2 is a latitude/longitude step; step 4 is an hour-of-day step; CZ_TYPE_ENC is in the top-4 of every step. The architecture is recognizable as feature-selecting in the way Arik and Pfister [2021] describe — but on this 8-feature task it never has enough irrelevant features to filter away. The local-attribution explain matrix, which weights per-step masks by each step’s contribution to the prediction, re-ranks the top two: mean |explain| puts DURATION_MIN (4.24) ahead of CZ_TYPE_ENC (2.88), because step 5’s heavy DURATION_MIN allocation carries large prediction weight despite a marginal mask share of only 17.7%. WFO_ENC is last under either ranking (0.94).

Cross-model comparison. Reduced to ranks on the common 8-feature axis, the two models agree much more than the distribution shapes suggest.

feature	XGB rank (gain)	TN rank (mask)	TN rank (explain)
CZ_TYPE_ENC	1	1	2
DURATION_MIN	2	2	1
STATE_ENC	3	7	7
MONTH	4	6	3
WFO_ENC	5	8	8
BEGIN_LON	6	5	5
HOURL	7	3	4
BEGIN_LAT	8	4	6

The two top features (CZ_TYPE_ENC, DURATION_MIN) are the top-two under all three rankings; the bottom feature (WFO_ENC) is in the bottom-two under all three. The middle ordering disagrees — XGBoost places STATE_ENC third but TabNet places it seventh; TabNet places HOURL and BEGIN_LAT in its top half but XGBoost places them seventh and eighth. The discrepancy reflects an architectural difference: XGBoost encodes location through STATE_ENC because boosted trees split repeatedly on that integer-coded category, while TabNet’s masks distribute location information across the raw lat/lon features instead. Both models are solving the same problem with the same information; they package the location signal differently.

The agreement at the top and bottom of the ranking is the load-bearing observation for §10. Both models converge on a feature subset of roughly two strong features and six weaker but non-negligible features. There is no “irrelevant” feature for TabNet’s attention to filter out — the worst feature on this task (WFO_ENC) still earns 4.3% of TabNet’s mask weight and 0.8% of XGBoost’s gain.

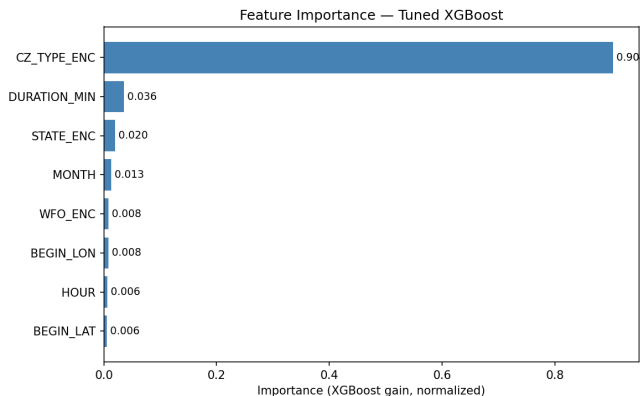


Figure 10: Tuned-XGBoost split-gain importance over the 8 engineered features. The visual emphasizes the 90%-on-CZ_TYPE_ENC concentration.

10 Discussion

The empirical sections leave three things to explain: why the random-split macro-F1 result came out against the pre-registered hypothesis, why the 2023→2024 temporal split partially recovers it on exactly the two named boundary classes, and what to do with the moderate cross-model error independence we measured in §8.2.

Why XGBoost wins the random split. TabNet’s architectural promise is sparse, instance-adaptive feature selection regularized by a sparsity prior γ [Arik and Pfister, 2021]. The mechanism is most valuable when *selection itself* is the hard part of the problem — when the input has many redundant or irrelevant features. On our 8-feature working set, selection is not the hard part: even the worst feature (WFO_ENC) receives 4.3% of TabNet’s mask weight and 0.8% of XGBoost’s gain, so there is no idle feature for attention to filter away. This is exactly the regime Shwartz-Ziv and Armon [2022] flag as least favorable for attention-based methods. In that regime the gradient-boosted baseline is at its strongest: with only 8 features XGBoost’s greedy gain-maximization is close to exhaustive. The negative

result is not that TabNet’s attention failed (§9 shows it learned coherent step-specialized masks) but that its comparative advantage requires a feature regime this dataset does not provide.

Why the temporal qualification is consistent with the mechanism. TabNet beats XGBoost on Hail (+0.037) and Thunderstorm Wind (+0.017) under 2023→2024 shift, degrading 3–4× less than XGBoost on those classes. The §7 audit makes a mechanistic reading available: TabNet produces softer, less-confident decision boundaries (§7.1, §7.3). A tight, sharply-confident fit to 2023 pays the largest penalty under distribution shift in boundary regions, which is where the two proposal-named classes live; XGBoost’s random-to-temporal drops (−0.207 Hail, −0.120 TS Wind) dwarf TabNet’s (−0.057, −0.045). We offer this as a hypothesis of mechanism, not a proven claim — one annual fold is too thin (§11).

Cross-model error independence and the ensemble follow-up. §8.2 measured Cohen’s $\kappa = 0.47$: 3,191 of 23,544 test rows are correctable by exactly one of the two models. A confidence-weighted ensemble would in principle close some of that gap, with Hail (790 both-wrong) the smallest-headroom class. We do not run that experiment here; it is the most defensible follow-up.

11 Limitations

Four limitations bound the scope of the claim. **(i) Feature count:** our 8-feature engineered working set is the very regime in which TabNet’s selection mechanism has nothing to filter; the negative result does not transfer mechanically to wider tabular tasks where only a handful of many features carry signal. **(ii) Single-year temporal fold:** the 2023→2024 per-class flip on Hail and Thunderstorm Wind is suggestive but one annual fold cannot support a domain-adaptation claim; a leave-one-year-out design across multiple NOAA archives would be the right next step. **(iii) No ensemble baseline:** $\kappa = 0.47$ leaves room for a confidence-weighted ensemble, but we did not evaluate one. **(iv) ECE binning sensitivity:** we used 10 equal-width bins per Guo et al. [2017], fixed before the audit; the audit’s load-bearing numbers (Brier, NLL, post-temperature-scaling ECE) are not materially bin-dependent, which is part of why the audit is robust where ECE alone is not.

12 Conclusion

We pre-registered the hypothesis that TabNet’s sequential attention would outperform a tuned gradient-boosted baseline on NOAA Storm Events 2023 — and on the principal random-split, iid evaluation, the hypothesis is falsified along every axis we tested. Tuned XGBoost beats refined TabNet by +0.049 macro-F1 (95% bootstrap CI [+0.044, +0.054]; McNemar $\chi^2 = 591.6$, $p < 10^{-130}$), and the direction is reversed on exactly the two boundary classes the proposal singled out. A surface reading of Expected Calibration Error initially favored TabNet, but a three-control audit — proper scoring rules, temperature scaling, and argmax invariance — overturned that reading: XGBoost is strictly better under Brier and NLL, and a single learned scalar $T = 1.26$ makes XGBoost the better-calibrated model on ECE as well. The one nuance we report honestly is the 2023→2024 temporal fold, where TabNet’s softer fit degrades 3–4× less than XGBoost’s on Hail and Thunderstorm Wind, partially recovering the proposal’s per-class prediction. The broader methodological takeaway is the audit itself: ECE-only calibration claims in the tabular-DL literature should be paired with at least one proper scoring rule and a temperature-scaling baseline before they are believed.

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